HAND: ELECTRONICS HANDBOOK

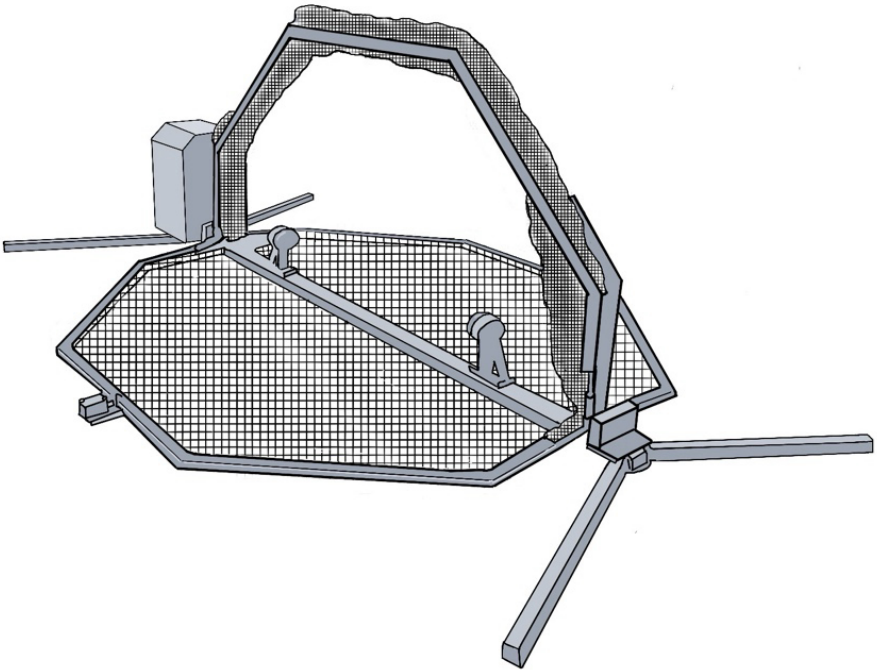


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Warning:

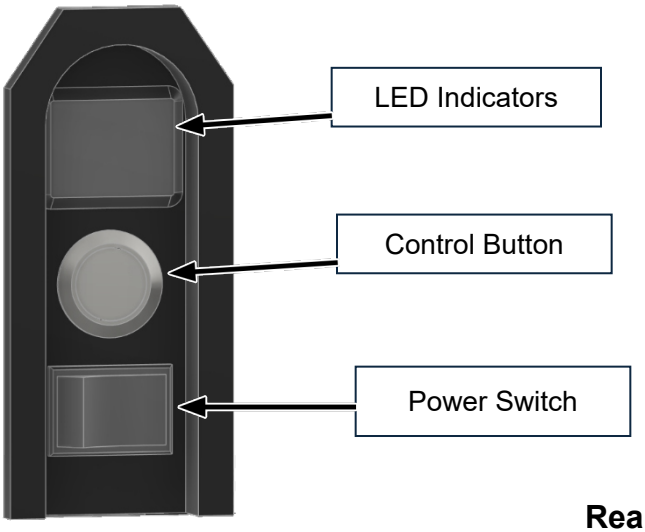
Always remove or securely fasten the battery prior to transporting the HAND.

Always keep battery terminals insulated from conductive materials.

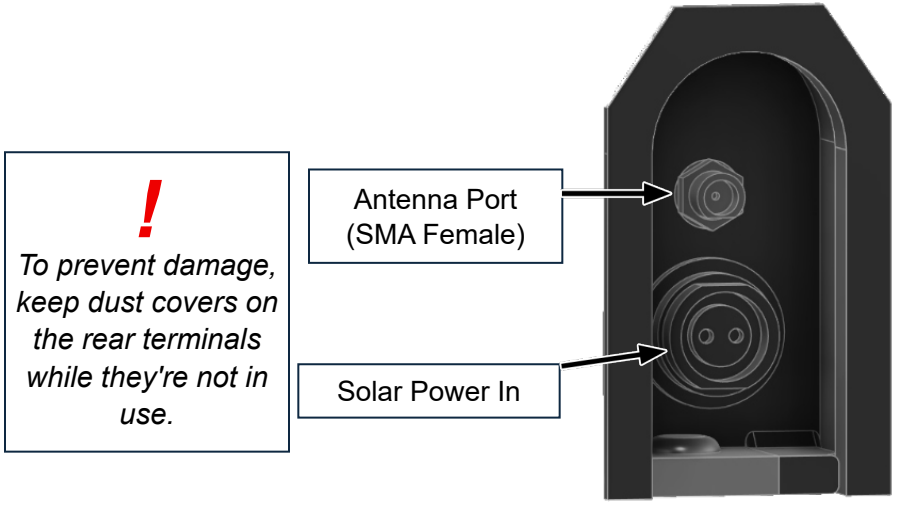
Keep hands away from all mechanical components while the trap is armed.

Inputs and Controls

Front Interface:



Rear Terminals:



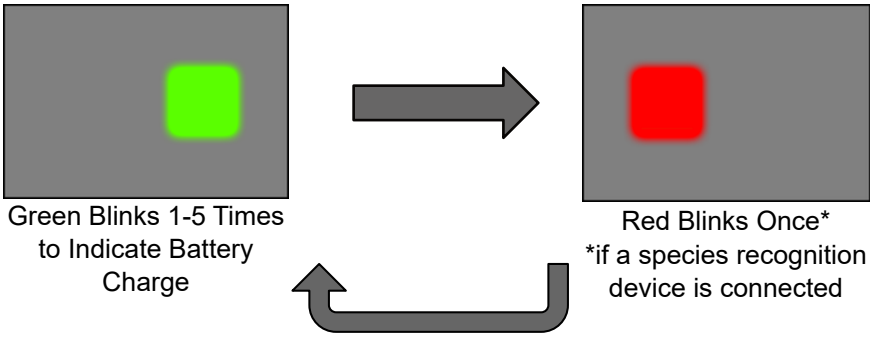
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To prevent damage, keep dust covers on the rear terminals while they're not in use.

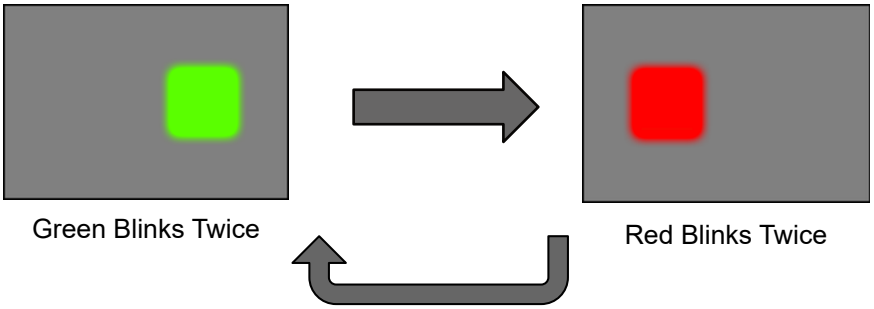
Indicators:

- Switch on using the power switch
- LEDs will illuminate to show the trap's current state, e.g.

Normal Mode:



Config Mode:



- The LEDs will turn off after 2 minutes. Short press the control button once to illuminate them again.

When Triggered:

If the HAND has been triggered, both LEDs will flash rapidly.

To **reset** back to operating mode, hold the control button for 3 seconds until the red LED is solid.

Quick Start Guide:

1. Check if the HAND is in config mode. If it is not in config mode, press and hold the control button for 10 seconds until there is a solid green LED then release.
2. Connect any internet-capable device (e.g. phone or laptop) to the “HAND_0_Config” Wi-Fi network, ensure the device is not connected to any other networks, including cellular.
3. If the configuration page doesn’t automatically show, enter “hand.config” or “192.168.4.1” into any web browser address bar on your device.
4. In the configuration page, it is recommended to change the device number to be unique from all other HANDs being deployed on your site. If the unit is new from factory, a firmware update is also recommended.
5. If an external antenna is connected check the “use external antenna” box. If required, see the “Config Mode” (p.7) section for a more detailed explanation of other available settings.
6. When done, press “save settings” and “exit config mode”. The HAND will disconnect and reboot into “standard mode”.

Standard Mode:

While disarmed in standard mode the LEDs indicate the trap state:

- 1-5 short green blinks indicate the battery charge level; 1 being the lowest and 5 being the highest.
- The red LED blinks when a Bluetooth camera is connected.

Manual Arming:

Double press the control button to manually arm the trap. The red LED will constantly flash to indicate it's armed.

Once manually armed the HAND will not disarm until it is told to by an external device or the control button is double pressed again.

Other Actions:

- To **reboot**, hold the control button for 3 seconds until the red LED is solid.
- To **enter config mode**, hold the control button for 10 seconds until there is a solid green LED, then release.
- To **factory reset** all settings to default, hold the control button for 20 seconds until both LEDs blink rapidly.

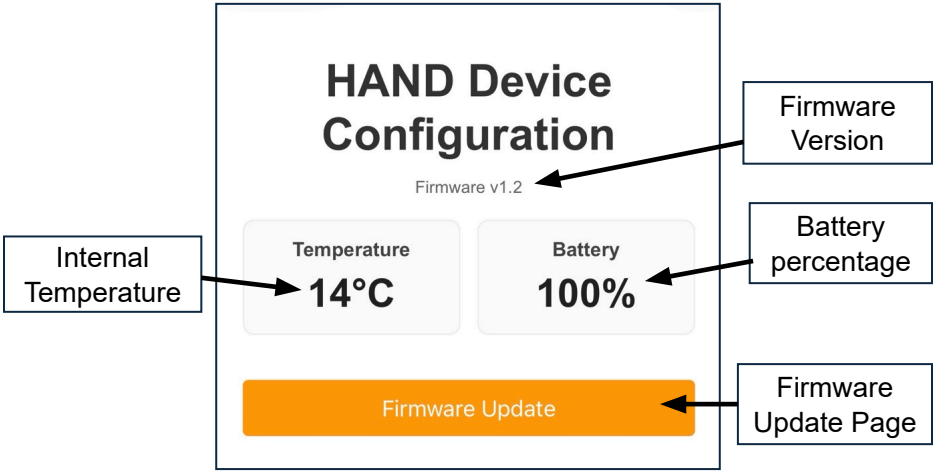
Wi-Fi Credentials:
-SSID: HAND_#_CONFIG
*where '#' is the HAND number.

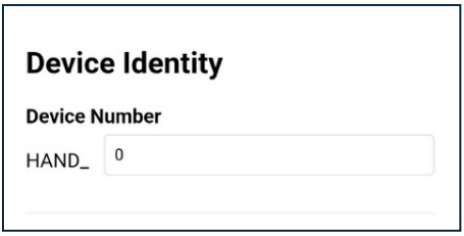
Config Mode:

While in config mode, the HAND creates its own Wi-Fi hotspot and camera connections are disabled. The Wi-Fi hotspot is used to access a config page where features and settings can be changed. The HAND will automatically revert back to operating mode after 10 minutes of inactivity in config mode.

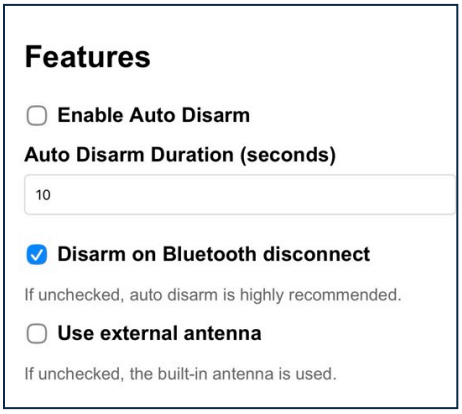
Accessing the Config page:

- Stay close to the HAND.
- Connect to the Wi-Fi hotspot using a phone or computer.
- In a web browser put **hand.config** or **192.168.4.1** in the address bar.
- A screen like the following will show:





Device Number- The number the HAND will advertise for device pairing. This should be made unique to any other HAND's being deployed on site. *0 by default.*

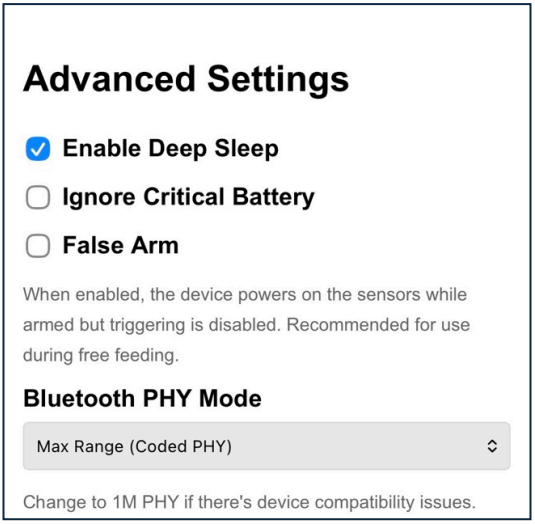


Auto Disarm- The HAND can be set to automatically disarm a chosen time period after being armed. *OFF by default.*

Disarm on Bluetooth disconnect- Controls whether the HAND disarms upon losing connection to the paired species recognition device while armed.

Use external antenna- turn this on if an external antenna is plugged into the SMA port. External antennas can extend the HAND's effective wireless range. *OFF by default.*

WARNING: Do not use an antenna that exceeds +2dBi

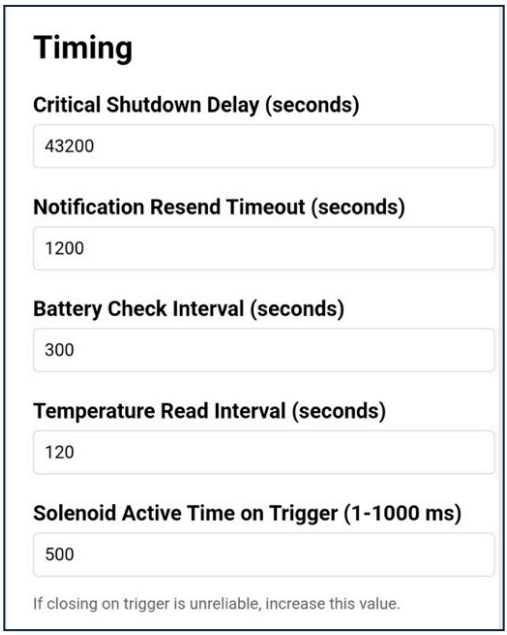


Enable deep sleep- When enabled, the HAND will enter a “deep sleep” state to conserve battery after being triggered. Once in deep sleep, all wireless communications cease and the HAND can only be woken from a button press or power cycle. *ON by default.*

Ignore Critical Battery- While the battery is in a critically low state of charge, the HAND will not arm or trigger. This behavior can be turned off by enabling this setting, but this may cause damage to the battery. *OFF by default.*

False Arm- While enabled the HAND will arm/disarm as normal but will not trigger. Best for use during free feeding periods, allowing the target to get more accustomed to the sensors.

Bluetooth PHY Mode- Determines how Bluetooth transmits wireless signals. Coded PHY uses special techniques to achieve maximum range, but may be unreliable if a connected device does not fully support the Bluetooth 5.0 spec. Switch to 1M PHY if the connection with a species recognition device is unreliable.



Critical shutdown delay- The amount of time the HAND waits before turning off when the battery is critical (seconds). *43200 by default.*

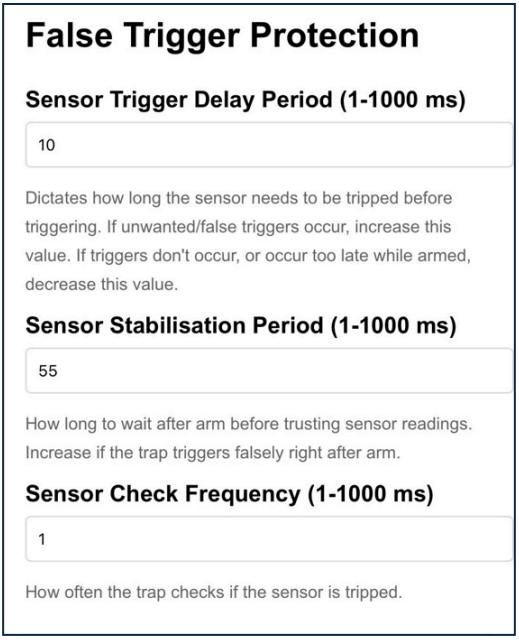
Notification Resend Timeout- When a major event occurs, such as a trigger, the HAND will continuously notify a connected device until it receives an acknowledgment or it times out (seconds). *1200 by default.*

Battery check interval- How often the HAND checks the battery charge in seconds. *300 by default.*

Temperature read interval- How often the HAND checks its internal temperature in seconds. *120 by default.*

Solenoid Active Time on Trigger- The length of the solenoid pulse when triggered (ms). Increasing this value may help if the hoops start closing unreliably, though other troubleshooting steps should be attempted first. *500 by default.*

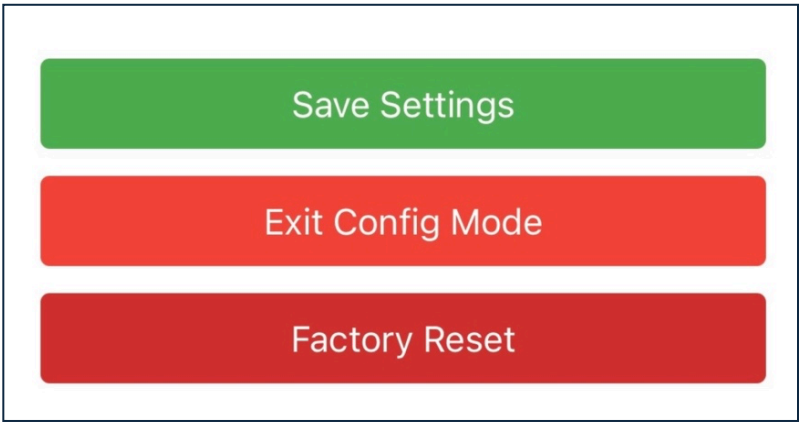
False Trigger Protection:



Sensor Trigger Delay Period- Dictates how long something must be between the beam sensors before the trap is triggered. A longer delay helps prevent false triggers on fast moving debris/insects while a shorter delay increases the responsiveness. It is recommended to test fire the trap after changing this value. *10 by default.*

Sensor Stabilisation Period- How long the sensors have to stabilise on arm before they can trigger the trap. Increase this if the trap is triggering immediately on arm with nothing obstructing the sensors. *55 by default.*

Sensor Check Frequency- Controls how often the device checks the sensor for a trigger. Higher values may conserve battery during long arm periods but will decrease trigger responsiveness. *1 by default.*



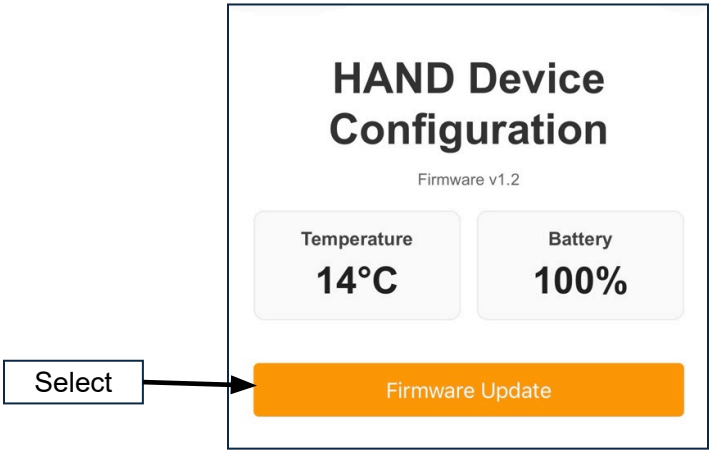
Save Settings- Press to apply any changed settings.

Exit Config Mode- Press to switch back to standard mode.

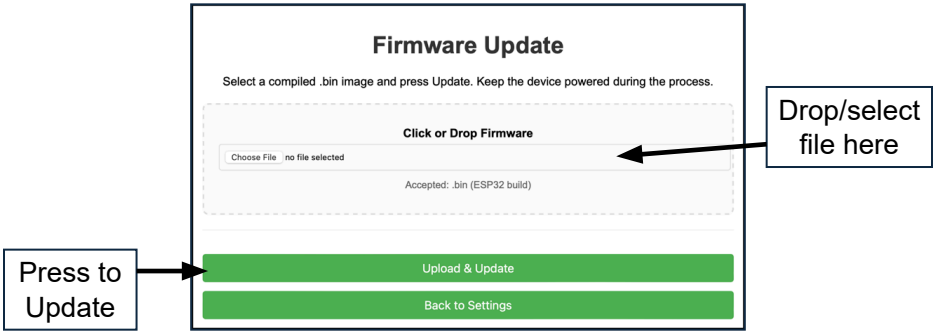
Factory Reset- Changes all settings back to default.

Firmware Updates:

- Download the latest firmware from here or the QR code.
- In the config page, select “Firmware update”:



- Choose or drop a ‘HAND_FW_#.#.#.bin’ file into the page.
- Press ‘Upload & Update’:



After the update completes, exit config mode by holding the config button for 3 seconds until there is a solid red LED or press ‘exit config mode’ on the web page.



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